



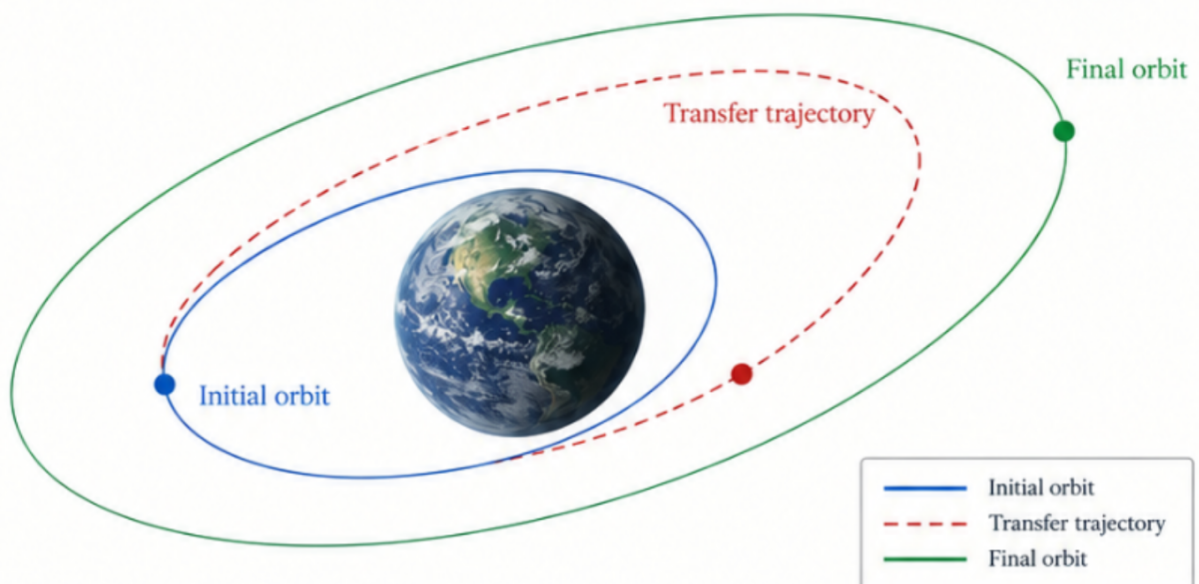
POLITECNICO
MILANO 1863

SCHOOL OF INDUSTRIAL
AND INFORMATION ENGINEERING
BSc in Aerospace Engineering

ACADEMIC PROJECT REPORT

Orbital Transfer Mission Analysis

Analysis and comparison of orbital transfer strategies
using numerical methods



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1. Introduction

The aim of the assignment is to transfer a satellite from an initial orbit to a final one, trying to study each possibility to spend less fuel (less specific impulse). The team performed studying individually each strategy purpose and shared their own result to the rest of the team. After a detailed examination, each of us filled in the report with all the data of the transfer strategy he worked on. We compiled introduction, initial and final orbit characterisation and conclusion. The presentation is the result of our team work.

2. Initial orbit characterisation

2.1.

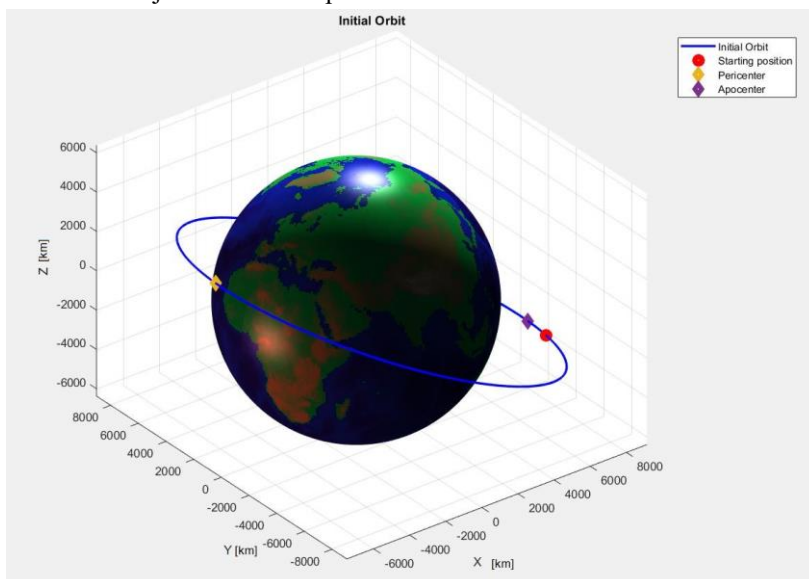
From the data received (initial position and its velocity), through the 'car2kep' Matlab function, we calculated the orbital parameters of the initial orbit.

Initial Position [km]	Velocity [km/s]
8925.0023	1.2740
$r1 = [-2106.9873]$	$v1 = [5.8400]$
-3959.8138	-0.5030

a1 [km]	e1 [-]	i1 [rad]	$\Omega1$ [rad]	$\omega1$ [rad]	$\Theta1$ [rad]
9094.6	0.0999	0.4156	1.5478	1.5392	2.9816

2.2.

As the initial orbit's eccentricity parameter is extremely low, the orbit is almost circular and it is situated in MEO. Initial position is located just before the apocenter.



2.3. Graphical representation of the initial orbit

3. Final orbit characterisation

3.1.

From the data received (orbital parameters), through the ‘kep2car’ Matlab function, we calculated the final position and its velocity.

a5 [km]	e5 [-]	i5 [rad]	Ω_5 [rad]	ω_5 [rad]	Θ_5 [rad]
14140	0.3731	1.1860	1.7280	2.9420	0.1652

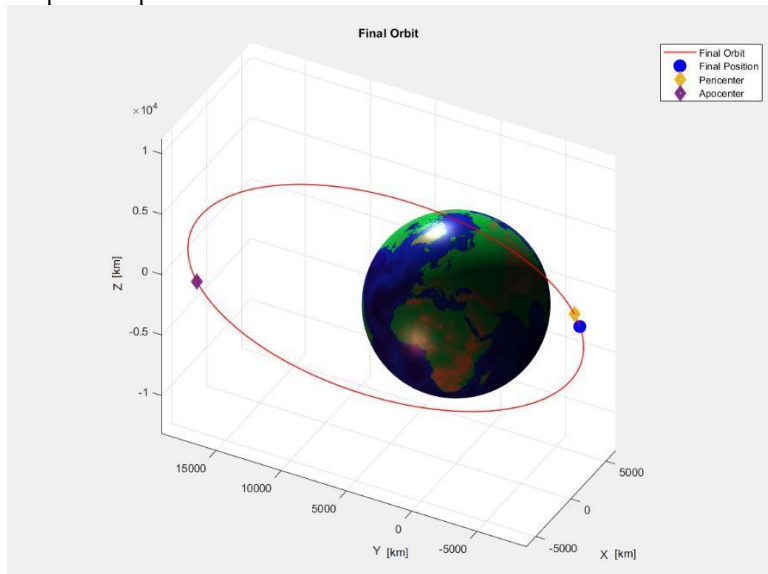
Initial Position [km]	Velocity [km/s]
1278.6825 r5=[-8800.3496 283.5688	2.9932 v5=[-0.1533 -7.2406

3.2.

The final orbit is bigger and more eccentric than the initial one. It’s also situated in MEO. Final position is situated shortly after the pericentre.

3.3.

Graphical representation of the final orbit.



4. Transfer trajectory definition and analysis

4.1.

The solution to reach the final position is to apply different transfer strategies:

- Standard strategy: it consists of a change of orbital plane, a change of the periapsis argument and a bitangent transfer to change the orbital shape studied in both apsidal points.
- Alternative standard strategy: it consists of a bitangent transfer to change the orbital shape studied in both apsidal points followed by a change of orbital plane and a change of the periapsis argument.
- Secant strategy (1): it consists of a secant transfer to change the shape and the periapsis argument of the orbit followed by a change of orbital plane.
- Secant strategy (2): it consists of a change of orbital plane followed by a secant transfer to change the shape and the periapsis argument.
- Alternative bielliptic strategy: it consists of a change of the periapsis argument followed by a bielliptic transfer with change of orbital plane in the apogee and another change of periapsis argument.

Standard Strategy

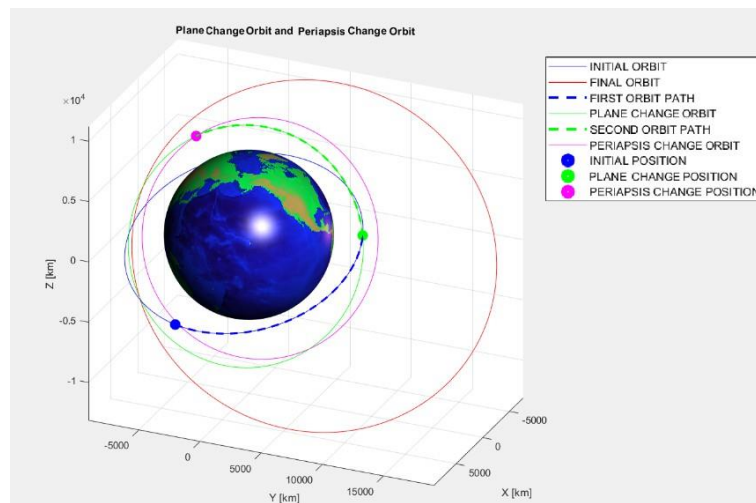
The standard strategy is composed by three manoeuvres:

The first one is an orbital plane change: it puts us from the initial orbit to another orbit (2) that has the same shape of the first one, but it changes its RAAN and its inclination. This type of manoeuvre also changes the periapsis argument of the orbit that is different from both the initial and final orbit. We make this transfer possible implementing in Matlab the 'changeOrbitalPlane' function.

a2 [km]	e2 [-]	i2 [rad]	Ω_2 [rad]	ω_2 [rad]
9094.6	0.0999	1.1860	1.7280	1.4037

The second manoeuvre is a change of the periapsis argument: it brings us from the second orbit to a different one (3) that has the same shape, inclination and RAAN of it, but with the periapsis argument like the final orbit. This transfer is applied in Matlab with the 'changePeriapsisArg' function.

a3 [km]	e3 [-]	i3 [rad]	Ω_3 [rad]	ω_3 [rad]
9094.6	0.0999	1.1860	1.7280	2.9420



The last manoeuvre is a bitangent transfer that put us in the final orbit.

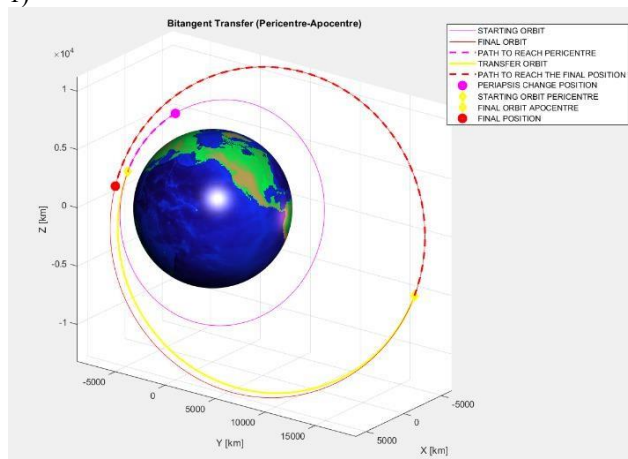
This manoeuvre can be studied in the two different apsidal point:

- 1) starting from the pericentre of the third orbit to the apocentre of the final one;
- 2) starting from the apocentre of the third orbit to the pericentre of the final one;

Both consist in a two-pulse manoeuvres that, by an elliptic transfer arc, moves us from the apsidal point of the third orbit to the opposite apsidal point of the final orbit.

For each manoeuvre, our satellite must reach an ideal point that make the transfer possible.

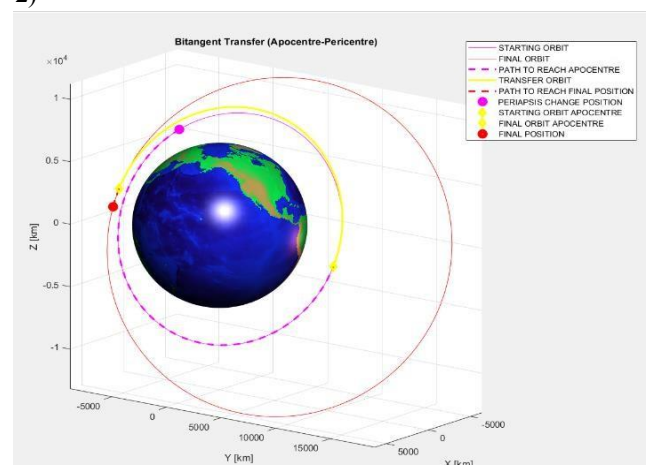
1)



Elliptic Tra Elliptic Transfer Orbit (1)

at1 [km]	et1 [-]	i3 [rad]	OM3 [rad]	om3 [rad]
13800.84	0.4068	1.1860	1.7280	2.9420

2)



Elliptic Transfer Orbit (2)

at2 [km]	et2 [-]	i3 [rad]	OM3 [rad]	om3 [rad]
9433.75	0.06035	1.1860	1.7280	2.9420

The entire strategy needs:

- 1) a total specific impulse of 7.1684 km/s and takes 22955 seconds (more than 6 hours) to reach the final point from the initial orbit;
- 2) a total specific impulse of 7.1949 km/s and takes 15396 seconds (more than 4 hours) to reach the final point from the initial orbit;

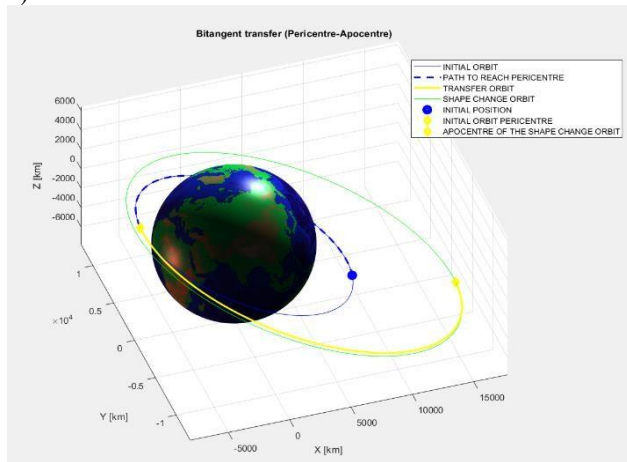
The second strategy, despite the more specific impulse required, even if it is very low, is the best strategy because it lasts 7559 seconds less (approximately 2 hour).

Alternative Standard Strategy

The alternative standard strategy makes the same manoeuvres of the standard strategy but in the opposite way. In fact, the first manoeuvre is a bitangent transfer that puts in us in a new orbit (2) with the same RAAN, periaapsis argument and inclination of the initial orbit but with the same shape of the final one. This manoeuvre can be done in the two apsidal point as we studied before 1), 2).

a2 [km]	e2 [-]	i2 [rad]	Ω 2 [rad]	ω 2 [rad]
14140	0.3731	0.4156	1.5478	1.5392

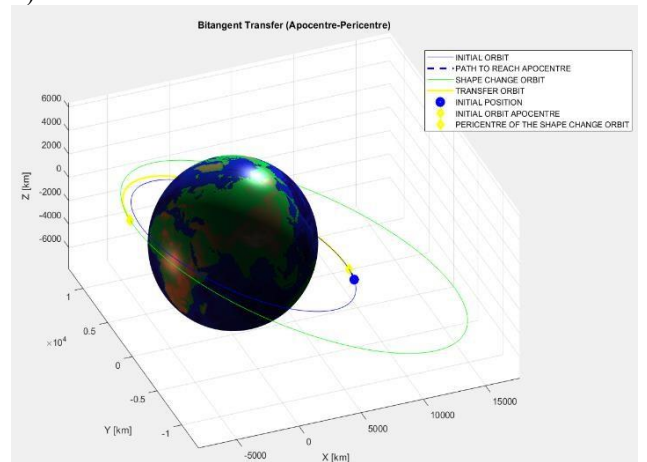
1)



Elliptic Tra Elliptic Transfer Orbit (1)

at1 [km]	et1 [-]	i3 [rad]	Ω 3 [rad]	om3 [rad]
13800.84	0.4068	0.4157	1.5478	1.5393

2)

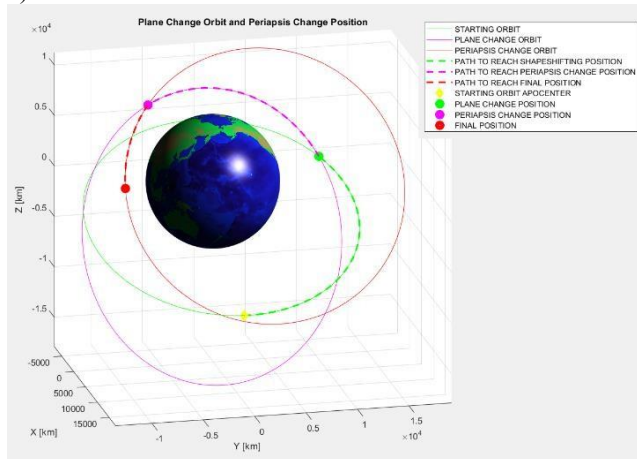


Elliptic Transfer Orbit (2)

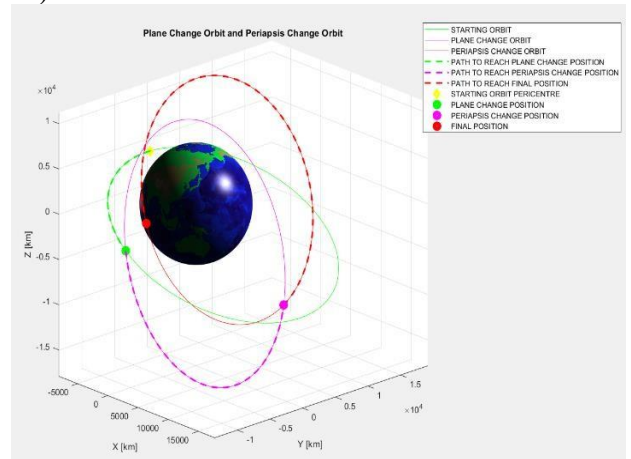
at2 [km]	et2 [-]	i3 [rad]	Ω 3 [rad]	ω 3 [rad]
9433.75	0.06035	0.4157	1.5478	1.5393

As opposed to the standard strategy, in this case the periaapsis argument and the plane of the orbit will change in different position to reach the final orbit and its final position. In fact, depending to the bitangent transfer choice, the strategy will be more efficient in terms of time or in term of specific impulse.

1)



2)



The entire strategy needs:

- 1) a total specific impulse of 8.8049 km/s and takes 23036 seconds (more than 6 hours) to reach the final point from the initial orbit;
- 2) a total specific impulse of 7.9658 km/s and takes 29247 seconds (more than 8 hours) to reach the final point from the initial orbit;

This strategy, as compared to the standard strategy, is never cheaper or faster.

Secant Strategy (1)

The Secant (1) strategy consist in 2 manoeuvres.

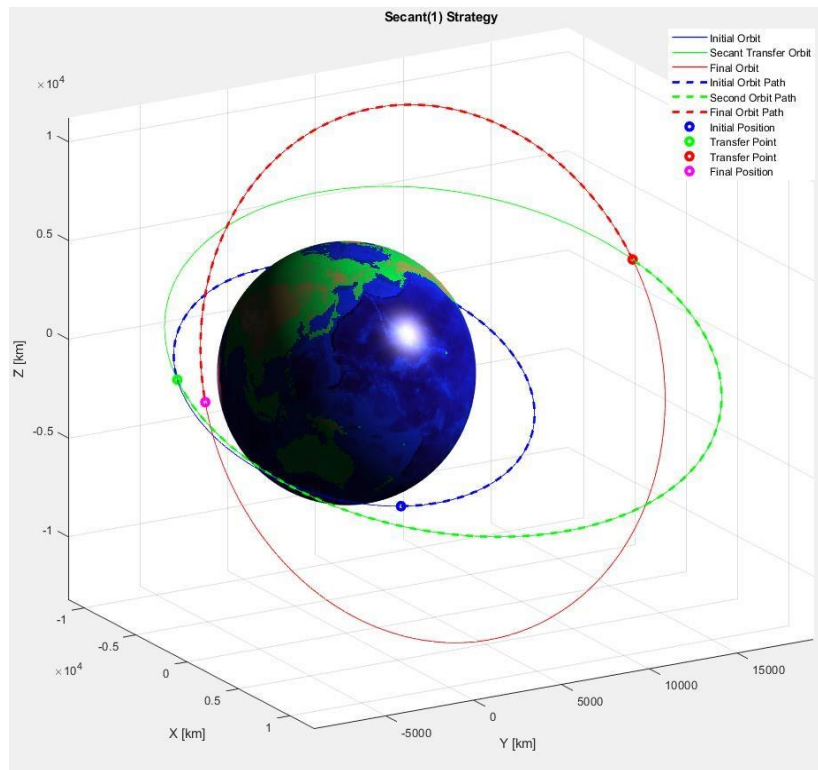
First of all, we calculated the pericentre anomaly necessary to obtain the same periapsis of the final orbit after the plane change.

The first manoeuvre is a Secant transfer. It goes from the initial orbit to an orbit with the same shape of the final orbit and the pericentre anomaly calculated before (2). The intersection point is determined using the Newton method in the function “changePeriapsisandShape”.

a2 [km]	e2 [-]	i2 [rad]	Ω_2 [rad]	ω_2 [rad]
14140	0.3731	0.4157	1.5478	3.0775

The second manoeuvre is a plane changing. It goes from the orbit reached before directly to the final orbit using the function “changeOrbitalPlane”. This is possible because of the precise periapsis change of the previous manoeuvre. Then we only have to wait until we reach the final position.

This strategy needs a total specific impulse of 4.2116 km/s and takes 23321.1 seconds (almost 6 and a half hour) to complete. This is the most efficient strategy studied in terms of impulse, mainly because the plane changing transfer happens at a bigger distance than in the Secant (2) strategy.



Secant Strategy (2)

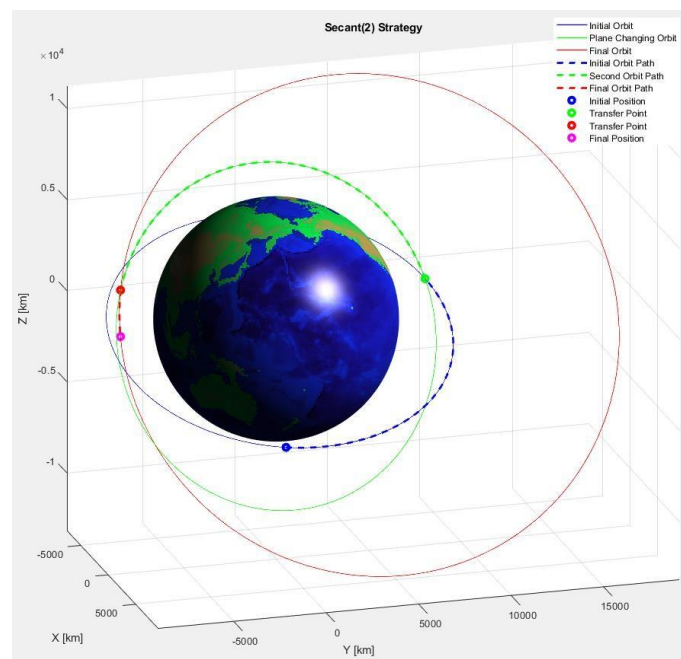
The Secant (2) strategy consists in two manoeuvres.

The first manoeuvre is a plane changing. It takes us from the initial orbit to an orbit with the same inclination and RAAN of the final orbit using the function “changeOrbitalPlane”. This transfer happens in first intersection and does not change the shape.

a2 [km]	e2 [-]	i2 [rad]	Ω 2 [rad]	ω 2 [rad]
9094.62	0.09999	1.1860	1.7280	1.4038

The second manoeuvre is a secant transfer: it happens in the first intersection point between the orbit in which we just transfer and the final orbit. It turns the shape and the pericentre anomaly in those of the final orbit using the function “changePeriapsisandShape”. After this transfer we only have to wait until we reach the final position.

The Secant (2) strategy needs a total specific impulse of 6.601 km/s and a total time of 6587.7 seconds (almost 110 minutes). This is the most efficient transfer strategy studied in terms of time, mainly because of the short path we have to travel in order to get in every transfer point.



Bielliptic Strategy

The last strategy we studied is the bielliptic one, and that’s because with the pericentre radius of the initial and final orbit we have, a bitangent strategy is always better than a bielliptic ($\frac{r_{p2}}{r_{p1}} < 11.94$). We decided to try anyway and the best way we find is to use a plane change during the bielliptic transfer.

First of all, we have to calculate the exact pericentre anomaly to do the plane changing exactly at the apogee of the two bielliptic arcs. The first manoeuvre is a periapsis change. It takes us to an orbit with the same shape and plane of the initial one, but with the pericentre anomaly calculated before, using the function “changePeriapsisArg”.

	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]
2 nd Orbit	9094.62	0.09999	0.4157	1.5478	3.3803

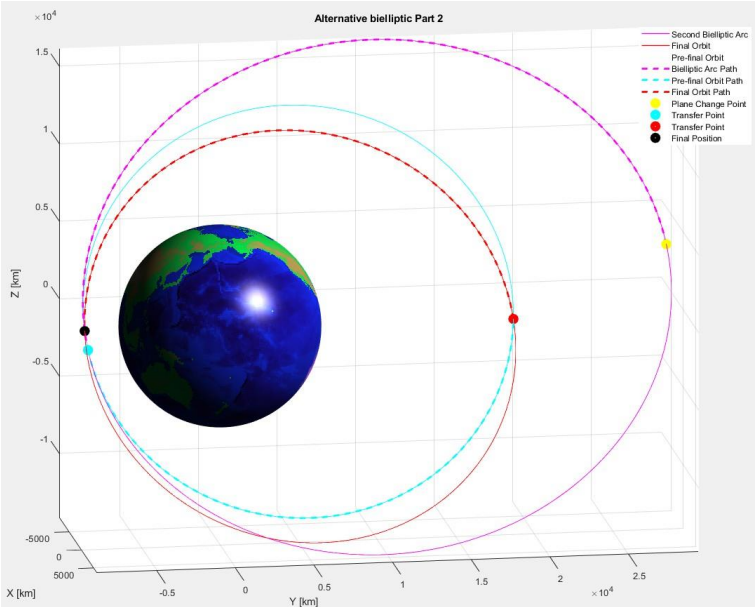
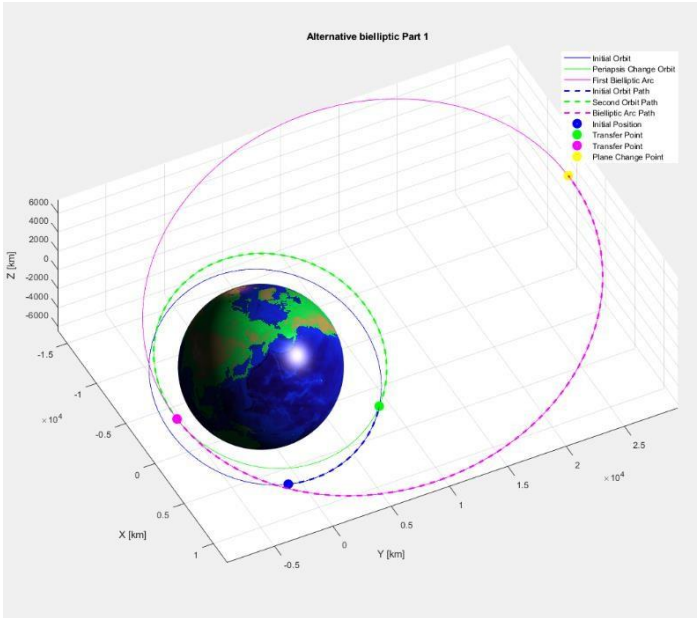
Then we start the bielliptic transfer from the second orbit’s perigee. When we arrive at the apogee of the first bielliptic arc, we start a bitangent transfer mixed with a plane changing. After that we are on the second bielliptic arc that has same inclination, RAAN and perigee radius of the final orbit.

1 st biell.arc	a [km]	e [-]	i [rad]	Ω [rad]	ω [rad]
2 nd biell.arc	19093.16	0.5712	0.4157	1.5478	3.3803
	19432.18	0.5438	1.1860	1.7280	3.2448

At the perigee of this arc, using a bitangent manoeuvre, we go into an orbit with same shape and plane of the final one. All this process is done by the function “BiellipticalPlaneChanging”. The dv in the plane changing is calculated using the Carnot’s Theorem.

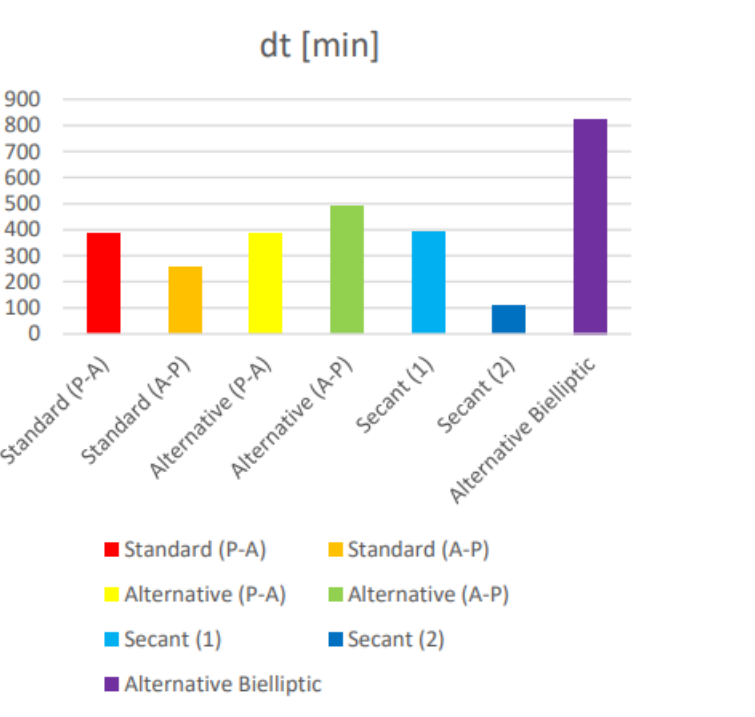
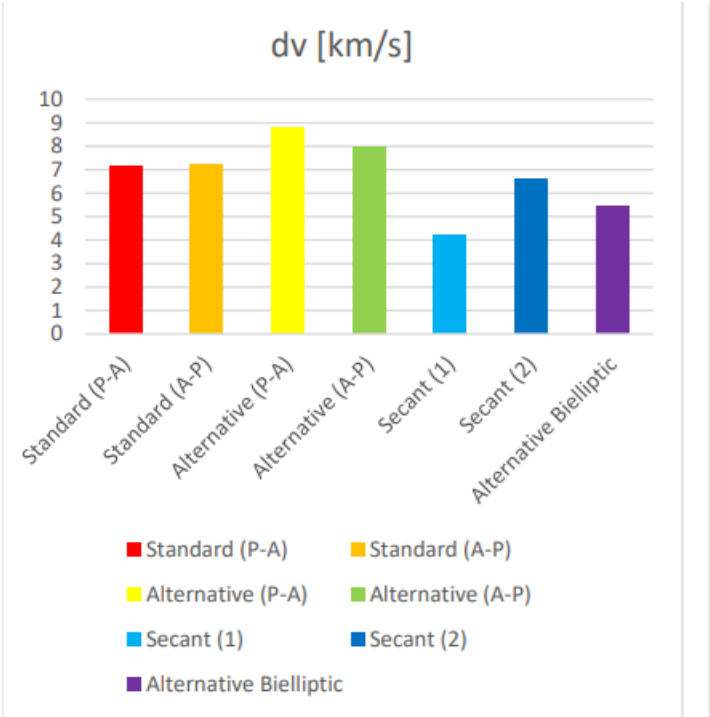
5 th Orbit	14140	0.3731	1.1860	1.7280	3.2448
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The last manoeuvre to do is another periapsis change that take us on the final orbit.



The apocenter radius for the bielliptic transfer represented is 30000 km. This strategy is very inefficient in time (49465 s or almost 13 h and 45 minutes) and not particularly efficient in terms of impulse (5.4471 km/s). This is, as mentioned before, mainly because of the initial and final conditions. Analysing the performances with different apocenter radius we find out that the lowest total impulse is found for an apocenter radius equal to the apocenter radius of the final orbit, the correspondent of Hohmann’s transfer with plane changing.

5. Conclusions



Considering all the strategies studied before, the most convenient strategy is the Secant (2) transfer. In fact, increasing by the 50% the propellant mass, we reduce the transfer time to $\frac{1}{3}$.

By the way, depending on the type of mission, Secant (1) may be more efficient: for example, for small satellite that can be loaded by a little amount of propellant.

Standard strategies result to be quite balanced, but none of them is better than Secant (1), (2).

Alternative standard strategies result to be never convenient in time or specific impulse.

As we said before, bi-elliptic transfer strategy, that is meant to be more convenient in terms of specific impulse, is never cheaper than the Secant (1) strategy due to initial and final conditions.

6. Appendix

Table 1: Standard Strategy (1)

t (s)	a (km)	e (-)	i (deg)	Ω (deg)	ω (deg)	θ (deg)	Δv (m/s)
0	9094.6184	0.09987	23.8166	88.6841	88.1930	170.8330	
3055.0	9094.6184	0.09987	23.8166	88.6841	88.1930	285.4820	5187.6
	9094.6184	0.09987	67.9527	99.0071	80.4311	285.4820	
5458.3	9094.6184	0.09987	67.9527	99.0071	80.4311	44.0665	924.5
	9094.6184	0.09987	67.9527	99.0071	168.5641	315.9334	
6333.9	9094.6184	0.09987	67.9527	99.0071	168.5641	360	958.3
	13800.84	0.4068	67.9527	99.0071	168.5641	0	
14401.5	13800.84	0.4068	67.9527	99.0071	168.5641	180	97.8
	14140.000	0.3731	67.9527	99.0071	168.5641	180	
22955	14140.000	0.3731	67.9527	99.0071	168.5641	9.4652	

Table 2: Standard Strategy (2)

t (s)	a (km)	e (-)	i (deg)	Ω (deg)	ω (deg)	θ (deg)	Δv (m/s)
0	9094.6184	0.09987	23.8166	88.6841	88.1930	170.8330	
3055.0	9094.6184	0.09987	23.8166	88.6841	88.1930	285.4820	5187.6
	9094.6184	0.09987	67.9527	99.0071	80.4311	285.4820	
5458.3	9094.6184	0.09987	67.9527	99.0071	80.4311	44.0665	924.5
	9094.6184	0.09987	67.9527	99.0071	168.5641	315.9334	
10649.7	9094.6184	0.09987	67.9527	99.0071	168.5641	180	130.1
	9433.75	0.06035	67.9527	99.0071	168.5641	180	
15209.1	9433.75	0.06035	67.9527	99.0071	168.5641	360	952.6
	14140.000	0.3731	67.9527	99.0071	168.5641	360	
15396	14140.000	0.3731	67.9527	99.0071	168.5641	9.4652	

Table 3: Alternative Standard Strategy (1)

t (s)	a (km)	e (-)	i (deg)	Ω (deg)	ω (deg)	θ (deg)	Δv (m/s)
0	9094.6184	0.09987	23.8166	88.6841	88.1930	170.8330	
4582.7	9094.6184	0.09987	23.8166	88.6841	88.1930	360	958.3
	13800.84	0.4068	23.8166	88.6841	88.1930	0	
12650.2	13800.84	0.4068	23.8166	88.6841	88.1930	180	97.8
	14140.000	0.3731	23.8166	88.6841	88.1930	180	
19296.6	14140.000	0.3731	23.8166	88.6841	88.1930	180	4778.6
	14140.000	0.3731	67.9527	99.0071	80.4311	285.4820	
21933.1	14140.000	0.3731	67.9527	99.0071	80.4311	44.0665	2969.9
	14140.000	0.3731	67.9527	99.0071	168.5641	315.9334	
23036.0	14140.000	0.3731	67.9527	99.0071	168.5641	9.4652	

Table 4: Alternative Standard Strategy (2)

t (s)	a (km)	e (-)	i (deg)	Ω (deg)	ω (deg)	θ (deg)	Δv (m/s)
0	9094.6184	0.09987	23.8166	88.6841	88.1930	170.8330	
267.0	9094.6184	0.09987	23.8166	88.6841	88.1930	180	130.1
	9433.75	0.06035	23.8166	88.6841	88.1930	180	
4826.3	9433.75	0.06035	23.8166	88.6841	88.1930	360	952.6
	14140.000	0.3731	23.8166	88.6841	88.1930	360	
7708.3	14140.000	0.3731	23.8166	88.6841	88.1930	360	3913.2

	<i>14140.000</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>80.4311</i>	<i>105.4820</i>	
16943.4	<i>14140.000</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>80.4311</i>	<i>224.0655</i>	<i>2969.9</i>
	<i>14140.000</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>168.5641</i>	<i>135.9334</i>	
29247.4	<i>14140.000</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>168.5641</i>	<i>9.4652</i>	

Table 3: Secant (1)

t (s)	a (km)	e (-)	i (deg)	Ω (deg)	ω (deg)	θ (deg)	Δv (m/s)
0	<i>9094.6184</i>	<i>0.09987</i>	<i>23.8166</i>	<i>88.6841</i>	<i>88.1930</i>	<i>170.8330</i>	
6277.6	<i>9094.6184</i>	<i>0.09987</i>	<i>23.8166</i>	<i>88.6841</i>	<i>88.1930</i>	<i>81.8833</i>	<i>1413.3</i>
	<i>14140</i>	<i>0.3731</i>	<i>23.8166</i>	<i>88.6841</i>	<i>176.3278</i>	<i>353.7491</i>	
16377.2	<i>14140</i>	<i>0.3731</i>	<i>23.8166</i>	<i>88.6841</i>	<i>176.3278</i>	<i>197.3476</i>	<i>2798.2</i>
	<i>14140</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>168.5642</i>	<i>197.3476</i>	
23321.1	<i>14140</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>168.5642</i>	<i>9.4653</i>	

Table 5: Secant (2)

t (s)	a (km)	e (-)	i (deg)	Ω (deg)	ω (deg)	θ (deg)	Δv (m/s)
0	<i>9094.6184</i>	<i>0.09987</i>	<i>23.8166</i>	<i>88.6841</i>	<i>88.1930</i>	<i>170.8330</i>	
3055.0	<i>9094.6184</i>	<i>0.09987</i>	<i>23.8166</i>	<i>88.6841</i>	<i>88.1930</i>	<i>285.4820</i>	<i>5187.6</i>
	<i>9094.6184</i>	<i>0.09987</i>	<i>67.9527</i>	<i>99.0071</i>	<i>80.4311</i>	<i>285.4820</i>	
6277.6	<i>9094.6184</i>	<i>0.09987</i>	<i>67.9527</i>	<i>99.0071</i>	<i>80.4311</i>	<i>81.8832</i>	<i>1413.3</i>
	<i>14140</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>168.5642</i>	<i>353.7490</i>	
6587.7	<i>14140</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>168.5642</i>	<i>9.4653</i>	

Table 6: Bielliptic Alternative

t (s)	a (km)	e (-)	i (deg)	Ω (deg)	ω (deg)	θ (deg)	Δv (m/s)
0	<i>9094.6184</i>	<i>0.09987</i>	<i>23.8166</i>	<i>88.6841</i>	<i>88.1930</i>	<i>170.8330</i>	
1760.0	<i>9094.6184</i>	<i>0.09987</i>	<i>23.8166</i>	<i>88.6841</i>	<i>88.1930</i>	<i>232.7409</i>	<i>1057.8</i>
	<i>9094.6184</i>	<i>0.09987</i>	<i>23.8166</i>	<i>88.6841</i>	<i>193.6749</i>	<i>127.2597</i>	
7568.8	<i>9094.6184</i>	<i>0.09987</i>	<i>23.8166</i>	<i>88.6841</i>	<i>193.6749</i>	<i>360</i>	<i>1428.7</i>
	<i>19093.2</i>	<i>0.5712</i>	<i>23.8166</i>	<i>88.6841</i>	<i>193.6749</i>	<i>0</i>	
20696.8	<i>19093.2</i>	<i>0.5712</i>	<i>23.8166</i>	<i>88.6841</i>	<i>193.6749</i>	<i>180</i>	<i>1842.4</i>
	<i>19432.2</i>	<i>0.5438</i>	<i>67.9527</i>	<i>99.0071</i>	<i>185.9133</i>	<i>180</i>	
34175.9	<i>19432.2</i>	<i>0.5438</i>	<i>67.9527</i>	<i>99.0071</i>	<i>185.9133</i>	<i>360</i>	<i>-474.2</i>
	<i>14140</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>185.9133</i>	<i>0</i>	
41727.0	<i>14140</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>185.9133</i>	<i>171.3262</i>	<i>644.0</i>
	<i>14140</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>168.5642</i>	<i>188.6738</i>	
49465.0	<i>14140</i>	<i>0.3731</i>	<i>67.9527</i>	<i>99.0071</i>	<i>168.5642</i>	<i>9.4653</i>	